

Gold Price Prediction ML

External Exam Preparation Guide

App: <https://gold-price-prediction-ml-f7cgv3jknks3vzds3esyz.streamlit.app/>

GitHub: <https://github.com/RammoKun/gold-price-prediction-ml>

1. What Does This Project Do?

A Streamlit web application that analyzes 25 years of gold price data (2000-2025), trains 4 ML models to predict gold prices, shows how major global events affect gold prices, lets users input custom values for real-time predictions + 6-month forecasts, and is deployed online with no installation required.

2. Dataset

- > **Source:** DataHub Core Gold Prices
- > **Frequency:** Monthly
- > **Period:** January 2000 - July 2025
- > **Total Records:** 307
- > **Target:** Gold Price (USD per troy ounce)

Events Analyzed

Event	Year	Type
9/11 Terror Attacks	2001	Geopolitical
Global Financial Crisis	2008	Economic
European Debt Crisis	2010	Economic
Oil Price Crash	2014	Economic
COVID-19 Pandemic	2020	Health Crisis
Russia-Ukraine War	2022	Geopolitical

Features Engineered

- > Lag variables: Price_Lag1, Price_Lag2, Price_Lag3 (previous months prices)
- > Moving averages: MA3, MA6 (3-month and 6-month rolling)
- > Cyclical encoding: Month_Sin, Month_Cos (captures seasonality)
- > Event dummies: Binary indicators for each event type
- > Year: Captures long-term upward trend

3. Machine Learning Models

3.1 Linear Regression

- > **What:** Statistical baseline - straight-line relationship
- > **Performance:** MAPE: 2.58%
- > **Limitation:** Fails during volatile/shock periods

3.2 Random Forest (200 trees)

- > **What:** Ensemble of 200 decision trees
- > **Performance:** MAPE: 11.67%
- > **Limitation:** Cannot predict beyond training range

3.3 XGBoost (200 estimators)

- > **What:** Gradient boosting, learning rate 0.05
- > **Performance:** MAPE: 12.46%
- > **Limitation:** Same extrapolation issue as RF

3.4 Neural Network (MLP)

- > **What:** 3 hidden layers 128-64-32, ReLU, Adam

- > **Performance:** MAPE: 6.10% - Best balance
- > **Advantage:** Can extrapolate beyond training range

Performance Summary

Model	RMSE	MAE	MAPE
Linear Regression	73.99	55.98	2.58%
Random Forest	519.79	307.98	11.67%
XGBoost	541.14	326.51	12.46%
Neural Network	212.56	148.52	6.10%

4. Training Methodology

- > **Chronological Split:** Train: 2000-2020 (252 obs), Test: 2021-2025 (55 obs)

Why chronological? In time-series, random split leaks future data into training. Chronological split evaluates models on truly unseen future data.

- > **Feature Scaling:** MinMaxScaler normalizes all features to [0,1] range
- > **Why not LSTM?:** Python 3.14 lacks TensorFlow. LSTM can run on Google Colab (future work).

5. App Architecture (4 Tabs)

Tab 1 - Data Overview

- > Gold price dataset table with stats (records, date range, max price)
- > Trend chart (2000-2025)
- > Full dataset viewer with CSV download

Tab 2 - Model Performance

- > Comparison table of all 4 models
- > Select any model to see actual vs predicted graph
- > Key insight text explaining best model

Tab 3 - Prediction

- > Input: Year, Month, last 3 prices, event type
- > Choose which ML model to use
- > Get predicted gold price instantly
- > 6-month iterative forecast with line chart

Tab 4 - Event Analysis

- > Event statistics (mean price, volatility, count)
- > Timeline with markers at major events
- > Data explorer for any numeric column
- > Volatility comparison bar chart

6. Key Findings

- > Gold prices spike 76% during wars (,200/oz for Russia-Ukraine)
- > 40% higher during health crises (,750/oz for COVID-19)
- > Normal periods: ,250/oz average
- > Neural Network is the recommended model (MAPE 6.10%)
- > Price_Lag1 is the most important feature

7. Technology Stack

Component	Technology	Purpose
Language	Python 3.14	Core development
Web Framework	Streamlit 1.58	Interactive UI
ML Models	scikit-learn	LR, RF, MLP
Boosting	XGBoost 3.3	XGBRegressor
Data	Pandas, NumPy	Processing
Visualization	Matplotlib 3.11	Charts
Deployment	Streamlit Cloud	Web hosting
Version Control	Git + GitHub	Code mgmt

8. Project Structure

```
gold_price_predictor/
  app.py           - Streamlit app (4 tabs)
  train_models.py  - Batch training script
  generate_report.py - PDF report generator
  requirements.txt - Dependencies
  src/
    data_loader.py - Loads/cleans data
    models.py      - ML training/eval
    visualization.py - Chart generation
    analysis.py    - Event analysis
  output/         - Graphs & metrics
```

9. Common Examiner Questions

Q: Why not LSTM / deep learning?

A: Python 3.14 does not support TensorFlow yet. The MLP (scikit-learn) was used instead. LSTM can be run on Google Colab - mentioned in Future Work chapter.

Q: Why is Linear Regression MAPE so low (2.58%)?

A: The test period (2021-2025) had a strong upward trend. Linear Regression captures linear trends well but would fail in volatile periods. The 2.58% is misleading - NN (6.10%) is more robust overall.

Q: Why chronological split instead of random?

A: Random split causes data leakage in time-series - the model sees future patterns during training. Chronological split evaluates on truly unseen future data, giving realistic performance estimates.

Q: What makes NN better than tree models?

A: Tree models predict the average of training target values in leaf nodes - they cannot output values beyond training range. Neural Networks use continuous activation functions that allow extrapolation.

Q: How does the 6-month forecast work?

A: Iterative forecasting - model predicts month 1. That prediction becomes Price_Lag1 for month 2. Process repeats 6 times, feeding each prediction forward as input for the next month.

Q: Where did the data come from?

A: DataHub Core Gold Prices dataset (open source). Monthly gold prices in USD/oz, Jan 2000 - Jul 2025.

Q: How is this different from Almira project?

A: That project recommends business models for e-commerce. This project predicts gold prices using 4 ML models with event impact analysis, deployed as interactive Streamlit app.

10. Limitations & Future Work

- > **Limitations:** Tree models cannot extrapolate; no real-time data; TensorFlow unsupported; monthly frequency only
- > **Future:** Live APIs, LSTM on Colab, sentiment analysis (NLP), more data sources, automated retraining